The higher-rank property-(T) outcome for $\mathrm{Aut}(F_n)$ is answered by arXiv:1812.03456

Literature-status note for arXiv:1712.07167

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Status. Explicit later-paper answer to the substantive higher-rank outcome: $\mathrm{Aut}(F_n)$ has property (T) for every $n \geq 5$. The later proof uses a stronger rank-five sum-of-squares certificate, so this note does not claim that the bare assertion "$\mathrm{Aut}(F_5)$ has property (T)" is by itself a black-box inductive hypothesis, nor that Bridson's separate quotient question is solved.

1 The source question

Kaluba, Nowak, and Ozawa prove in arXiv:1712.07167 that $\mathrm{Aut}(F_5)$ has Kazhdan's property (T) [1]. On PDF page 2 they ask whether this fact can be used to deduce property (T) for $\mathrm{Aut}(F_n)$ for every $n \geq 5$, as for higher-rank lattices. They explain that their attempted group-theoretic induction breaks at Bridson–Vogtmann Question 12: if the image of $\mathrm{Aut}(F_n)$ in a quotient of $\mathrm{Aut}(F_{n+1})$ is finite, must the quotient be finite?

Thus the passage contains two levels:

1. the substantive outcome that every higher-rank $\mathrm{Aut}(F_n)$ has property (T);
2. a stronger request for an abstract induction from the rank-five fact, via or around the quotient question.

2 The later higher-rank theorem

Kaluba, Kielak, and Nowak's follow-up arXiv:1812.03456v2 states as Theorem 1 on PDF page 2 [2]:

$$\boxed{\mathrm{Aut}(F_n) \text{ has property } (T) \text{ for every } n \geq 6.}$$

The sentence immediately following the theorem combines it with the source's $n = 5$ result and records the corollary

$$\mathrm{Aut}(F_n) \text{ has property } (T) \qquad (n \geq 5).$$

This completely answers the practical higher-rank existence question. The follow-up also proves a uniform Kazhdan-radius bound for the index-two groups $\mathrm{SAut}(F_n)$.

The mechanism is genuinely extrapolative but uses more data than abstract property (T). The authors decompose Laplacian expressions into pieces whose rank-m sum-of-squares decompositions can be symmetrized to every larger rank. They reduce the infinite family to a single finite rank-five sum-of-squares certificate. They explicitly say that the results and methods of the source are instrumental.

3 Scope limitation

Because that proof uses a particular certificate in the group algebra, it does not establish a general theorem of the form

$$\mathrm{Aut}(F_n) \text{ has } (T) \Longrightarrow \mathrm{Aut}(F_{n+1}) \text{ has } (T)$$

from the property alone. Nor does the cited follow-up claim to settle the Bridson–Vogtmann quotient problem. The precise disposition is therefore:

$$\boxed{\text{all } n \geq 5: \text{ answered affirmatively; bare black-box induction and quotient question: not claimed.}}$$

References

[1] M. Kaluba, P. W. Nowak, and N. Ozawa, $\mathrm{Aut}(\mathbb{F}_5)$ *has property* (T), arXiv:1712.07167 (2017).

[2] M. Kaluba, D. Kielak, and P. W. Nowak, *On property* (T) *for* $\mathrm{Aut}(F_n)$ *and* $\mathrm{SL}_n(\mathbb{Z})$, arXiv:1812.03456v2 (2021).